Nithin K

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.52
Sri Chaitanya Techno School	Class XII (CBSE)	2024	94.60%
Sri Chaitanya Techno School	Class X (CBSE)	2022	97.80%

SKILLS

Languages: C, Python, Verilog

Digital Design and EDA: Intel Quartus, testbench design & verification, LTSpice

Libraries and Tools: Git, Nvim, Numpy, MATLAB

PROJECTS

8-Point Pipelined FFT (Radix-2, DIT)[Github]

- Designed a custom 8-point Radix-2 FFT in Verilog using a **Decimation-In-Time** pipelined architecture with bit-reversal input buffering and fixed-point twiddle factor multiplication
- Built a self-checking testbench that automatically validated FFT outputs against expected results across 5 signal types (cosine, impulse, DC, complex exponential, two-tone)
- Achieved **148.55 MHz** Fmax using fewer ALUTs than a comparable non-pipelined design

Pipelined FIR Filter[Github]

- Implemented a **100-tap FIR** low-pass filter in Verilog across three pipelining architectures (Direct, Optimized, Genvar-based) targeting Intel Cyclone V FPGA
- Identified and resolved DSP resource exhaustion in the direct architecture by restructuring the multiply-accumulate pipeline, enabling successful synthesis within device constraints
- Used Verilog generate blocks to build a parameterized, scalable pipeline structure, achieving the lowest signal-alignment latency (1 clock cycle) among all three designs
- Verified hardware output against a MATLAB fixed-point reference model using a custom Python signal-alignment and error-analysis script

Fixed-Point Division using Newton-Raphson[Github]

- Designed and implemented a fixed-point **(Q16.16)** division algorithm using the Newton-Raphson iterative method in Verilog, replacing direct division with reciprocal approximation and multiplication
- Derived a closed-form optimal linear initial-guess approximation via minimax/equioscillation analysis, minimizing iteration count needed for convergence
- Built a synchronous FSM-based hardware implementation in Verilog, verified against a Python golden model across multiple test cases with <0.01% relative error

MINI PROJECTS

ML from Scratch Personal Project[Github]

- Implemented a feedforward neural network from scratch in NumPy with hand-derived backpropagation, achieving **98%+ validation accuracy** on digit classification
- Built and compared 5 linear classification methods (least squares, Fisher's LDA, perceptron, logistic regression via IRLS, GMM via EM), analyzing robustness to outliers and convergence behavior
- Developed a tabular Q-learning agent for a simulated MDP, studying convergence and the exploration-exploitation tradeoff under an epsilon-greedy policy

RELEVANT COURSES

Machine Learning(EE2802), Data Structures and Applications(ID2230), Digital Systems Theory and Lab(EE1201 & EE1501), Probability(EE1080), Signal Processing(EE2000), Semiconductor Device Fundamentals(EE2103)

Analog Electronics(EE2400)**,** Semiconductor Device Fundamentals(EE2104), Electronic Devices and Circuits Lab(EE2301), Circuits and Network Analysis(EE1101)

POSITIONS OF RESPONSIBILITY

Core Member of Quiz Club [*June 2025 - June 2026]*

- Selected to represent the institution at the **Interrobang Quiz Festival** hosted by NALSAR University of Law
- Collaborated with a 3-member team to analyze complex trivia, current affairs, and lateral-thinking questions
- Secured **5th** out of **25+** competing teams from across India

EXTRACURRICULAR

Web Development Certification from Academor *[November 2024]*

Completed a training program covering fundamental web technologies including HTML, CSS, NodeJS.

Student Volunteer for National Service Scheme (NSS) [*June 2024 - Present]*

Actively participated in community outreach programs, including **campus cleanups, walkathons, etc**.

Team Member at Inter-Department Sports Competition (DIESTA) *[October 2025]*

Represented the Department of Electrical Engineering in competitive Badminton tournaments.